

Research agent

Two agents search and screen the literature.
A person decides what is kept.



Unit 11: Individual project presentation
Intelligent Agents, University of Essex Online
October 2026



plan



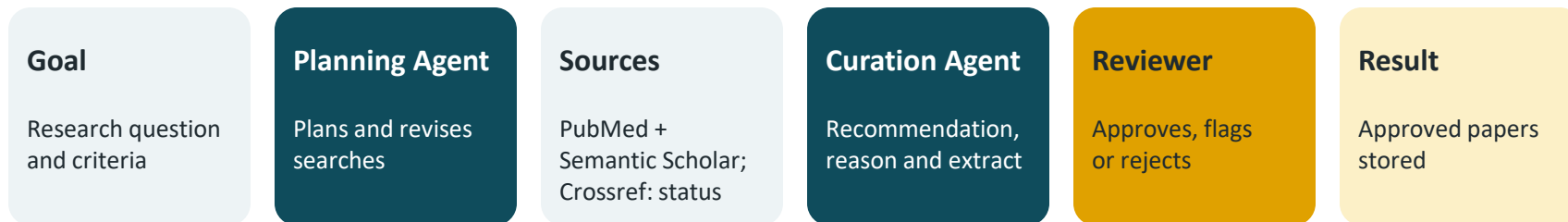
screen



decide



The design from Unit 6



Checked JSON handover; human approval before storage.



Selected classes and fields: request, shared state and run budget

BDI-inspired state: Rao and Georgeff (1995). Goal-based agent: Russell and Norvig (2022).

Action and observation loop: Yao et al. (2023).

Design to code: kept, changed, why

Kept

Two agents and a checked JSON handover

Human approval before storage

PubMed and Semantic Scholar;
Crossref for status

Gemini behind a replaceable adapter

SQLite, Streamlit and LangGraph

Changed

Full text omitted

Why

Unused source removed. Screening uses titles and abstracts.

18 model requests per run

Bounds each run. Daily account usage is not tracked.

Timeouts and counted retries

Busy or stalled requests led to bounded recovery.

Live run set to Flash-Lite

The eighth attempt completed. Flash remains the code default.



Changes and reasons recorded against the Unit 6 proposal.

The agent at work: a goal becomes a plan

Live goal, 18 September

Did the study evaluate ivermectin as a COVID-19 treatment and report clinical, virological or safety outcomes relevant to that treatment?

1 PubMed

ivermectin AND COVID-19 AND
(treatment OR clinical trial OR
randomized controlled trial OR outcome)

Finished: 50 papers

2 Semantic Scholar

ivermectin covid-19 treatment
clinical virological safety

Finished: 50 papers

Goal to plan to retrieval. After a search: continue, refine, drop or finish, with a reason.
Every model request counts towards the default run budget of 18.



Live goal and two completed searches. Run 98af89fa5cdc, 18 September.
Text transcribed from the submitted run log; these entries are not next-move decisions.

Retrieval, screening and the extract check



Efficacy and Safety of Ivermectin for Treatment and prophylaxis of COVID-19 Pandemic

rated 10 of 10 **flagged**

A. Elgazzar et al. (2020) · journal not given · DOI: [10.21203/rs.3.rs-100956/v1](https://doi.org/10.21203/rs.3.rs-100956/v1)

Publication status: No retraction or formal concern identified

Relevance: rated 10 of 10; the agent would include it

Extract from the paper (found in the paper):

Your note, kept until you decide again

Original treatment study, but withdrawn for data-integrity concerns

Live review excerpts, 18 September: withdrawn preprint, relevance 10/10; I flagged it.

Elgazzar et al. (2020); withdrawal: Research Square (2021).

approved

flagged

rejected

this paper cannot be approved: its extract was never looked for in the paper, so a stored claim could not be traced back to it. Flag it for a second judgement, or reject it

Invented paper: approval refused beside the decision

1 Retrieval

Matched by identifier, then title. 50 screening slots shared by turns across searches. Gaps stay visible.

2 Screening

Batches of ten. Include, exclude or uncertain, with a score, reason, summary and extract.

3 The check

The extract must occur in the title or abstract. Approval is blocked otherwise.

4 Guardrails

Paper text is marked as data. Prompt boundaries are tested; meaning still needs review.

Search reporting: Page et al. (2021); Rethlefsen et al. (2021)

The reviewer's gate, and what it cannot catch

Invented paper 1: a cohort study of drug X

Author 1 A (2016). Journal of Invented Studies. 10.99999/invented.1

Publication status: No retraction or formal concern identified

Relevance 10 of 10, the agent would include it

Why: an invented reason for paper 1, in one sentence

Summary: an invented summary of paper 1, built from its own wording

Extract (found in the paper): Symptoms improved in most of them by the end of the study.

Why, if you flag or reject it

Before, 16 September: extract and decision controls

7. Invented paper 7: a narrative review of drug X, with a title long enough to wrap onto several lines on a narrow screen and to show whether the identifier beside it stays readable rated 9 of 10

Author 7 A et al. (2022) - Journal of Invented Studies - DOI: 10.99999/invented.7

Publication status: No retraction or formal concern identified

Relevance: rated 9 of 10; the agent would include it

Reason: an invented reason for paper 7, in one sentence

Summary: an invented summary of paper 7, built from its own wording

Extract from the paper (found in the paper):

Symptoms improved in most of them by the end of the study.

Why, if you flag or reject it

After: a labelled card and separate extract

Observed

One participant: me.

Sessions on 16 and 19 September.

Changed and checked

Clearer cards and revision feedback.

Follow-up tasks: about 7–11 seconds.

Ease rating: 3 to 2; 1 is easiest.

Timings are not directly comparable.

Limit of the human gate

Lower-scoring papers can be hidden.

Approval controls what is stored;

it cannot recover papers never shown.





Evidence of execution

100

papers retrieved
50 from each search source

50

papers screened
50 places shared by turns

13

model requests
including revision

Live review

41 papers judged again; 4 recommendations changed.
Nine approvals matched nine stored identifiers.

CLEF: 50 records

Retrieval: 10/15, partial. Flash-Lite screening:
precision 6/6; relevant found 6/15. Nine relevant
papers were not recommended for inclusion.

CLEF-TAR (2019). Full methods and limits in the testing account.

Review complete: 9 papers approved

Screening of "Did the study evaluate ivermectin as a COVID-19 treatment and report clinical, virological or safety outcomes relevant to that treatment?", finished 18 Sep 2026, 20:00

All approved papers are saved

They are on this computer and are shown again from the start screen
under "Approved papers of an earlier run". Nothing is sent to anyone.

[Download the list](#)

9 approved - 0 flagged - 41 rejected

☐ View flagged and
rejected

Result: nine papers approved and saved

Development benchmark: 100 records

8/8 relevant retrieved at 1,000
requested results per search;
earlier run at 50: 0/8.
Known labels; separate runs.



Testing, and what it fixed

1,212

automated checks passed
3 skipped; 1 deselected
1 October 2026; Ruff clean

Why storage checks the extract again

```
# That check is made here rather than read off the record. `extract_found` says  
# what was true when the run ended, and a judgement can be edited afterwards:
```

```
if not is_in_the_paper(item.screening.extract, item.paper):  
    raise RefusedByStorage(  
        f"the extract now on {_naming(item.paper)} is not in the paper, "  
        "whatever the record of the earlier check says")
```

Comment and code excerpts, [processing/storage.py](#)

34 recorded remediation entries

R-21 Sensitive text escaped approval

Recursive rewriting and key checks

Six hostile approval tests

R-24 Invalid approval blocked the batch

Refusal beside the paper; storage guard kept

App and workflow tests

R-25 A model request kept waiting

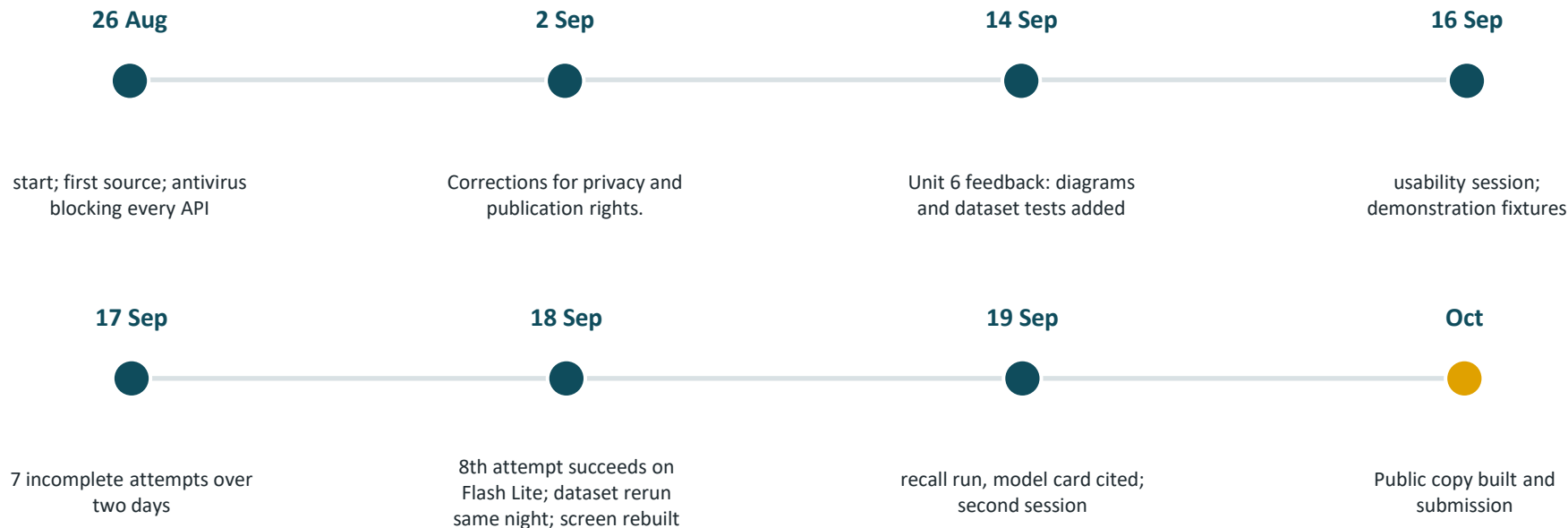
Connection timeout and counted retries

Timeout and request-count tests

Unit, integration and functional checks use invented replies. Live audits are separate.
Methods: [docs/testing.md](#). Decisions and remediation: [docs/development.md](#).



How the work developed over the module



The development document records the decisions, problems and corrections.

What the system achieves, and its limits

Conclusion

A working chain from a research question to searches, screening and stored approvals.

One completed live workflow, small datasets and one familiar participant.

Relevant papers can be missed or hidden.
Completeness is not established.

Next: a new topic with unseen labels, then testing with new reviewers.

Legal, social and ethical

UNESCO (2022) principles, mapped in the README.
Answered least well: fairness, on an uncalibrated ranking.

Key references

Rao and Georgeff (1995)

BDI agents: from theory to practice.
ICMAS-95, pp. 312–319.

Yao et al. (2023)

ReAct: synergizing reasoning and acting in language models. ICLR 2023.

Page et al. (2021)

The PRISMA 2020 statement.
International Journal of Surgery, 88, 105906.

Rethlefsen et al. (2021)

PRISMA-S. Journal of the Medical Library Association, 109(2), pp. 174–200.

Full references in the transcript.

Data: PubMed, Semantic Scholar, Crossref and CLEF TAR.
Software, models and AI acknowledgements: README.

